



MID SEMESTER EXAMINATION, AUTUMN-2024
School of Applied Sciences, KIIT-DU

B. Tech.
 1st Semester (2024 AB)
 Autumn 2024 (SAS)

Course: Basic Electronics

Code: EC10001

Full Marks: 20

Time: 90 minutes

Answer any FOUR QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

Q.1		Marks	CO
a)	A conductor and semiconductor is placed at room temperature. When temperature increases, write down the effect of temperature on the conductivity of the materials.	1	1
b)	A diode is biased with 0.9V so that current is flowing 20mA in it. Find the static resistance of the diode. Is it following ohm's law?	1	2
c)	What do you mean by depletion region in diode?	1	2
d)	Why Zener diode is used in voltage regulator?	1	2
e)	Draw the Symbol of NPN transistor with current direction.	1	3

Q.2		Marks	CO
a)	Explain drift current and diffusion current mechanism in semiconductor materials.	3	1
b)	A Si sample is doped with 'Boron' atoms so that hole concentration measured as $10^{18}/\text{cm}^3$. Find the minority carrier concentration at 300K? Assume intrinsic carrier concentration as $5 \times 10^{10} \text{cm}^{-3}$. <i>Si = 0.7 Boron = 3e P-type</i>	2	1

Q.3		Marks	CO
a)	In a center-tap full wave rectifier, forward resistance of each diode is 100Ω and input a.c with respect to each secondary winding is 10V (R.M.S). Assume load resistance of the circuit is $1K\Omega$ for the calculation. Solve for i) Peak voltage ii) Peak, Average, RMS value of load current iii) Efficiency.	3	2

b)	Draw the VI characteristics of PN junction diode with proper labeling.	2	2
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Q.4		Marks	CO
a)	Explain how Zener diode is used in a voltage regulator with proper circuit diagram.	3	2
b)	Write down four differences between semiconductor and conductor.	2	1

Q.5		Marks	CO
a)	Explain the different current components in PNP transistor with proper diagram.	3	3
b)	Write the relation between α and β for BJT. If $\alpha=0.995$ the find β .	2	3

$$I = I_s \left(e^{\frac{qV}{kT}} - 1 \right)$$

$$q = 1.6 \times 10^{-19}$$

$$k = 1.32 \times 10^{-23}$$

$$I_m.P. = (I_s)^2$$